

NanoGT Match Report: Hemophilia B

Disease: Hemophilia B (ORPHA:306)

Primary gene: F9

Gene CDS: 1383 bp

Inheritance: X-linked recessive

Target tissues: liver

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	DTX201	AAV8	9.6/10	High	phase2
2	Hemgenix	AAV5	9.5/10	High	approved
3	Roctavian	AAV5	9.5/10	High	approved
4	SPK-8011	AAVrh10	9.3/10	High	phase3
5	RGX-121	AAV9	8.3/10	High	phase3

Match #1: DTX201

Precedent disease: Hemophilia A

Vector: AAV8

Tissue target: liver

Composite score: 9.6 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Approval status: phase2

Match #2: Hemgenix

Precedent disease: Hemophilia B

Vector: AAV5

Tissue target: liver

Composite score: 9.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Approval status: approved

Match #3: Roctavian

Precedent disease: Hemophilia A

Vector: AAV5

Tissue target: liver

Composite score: 9.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)

- Pathway match: coagulation
- Approval status: approved

Match #4: SPK-8011

Precedent disease: Hemophilia A

Vector: AAVrh10

Tissue target: liver

Composite score: 9.3 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.5	2.0
Protein class	2.0	2.0
Pathway similarity	2.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Approval status: phase3

Match #5: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 8.3 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	0.5	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)

- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Different pathway (coagulation vs lysosomal_storage)
- Approval status: phase3